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Invited review

Hominin evolution in settings of strong environmental variability

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ABSTRACT

Investigations into how climate change shaped human evolution have begun to focus on environmental dynamics, i.e., the nature and tempo of climate and landscape variability, an approach that de-emphasizes static reconstructions of early hominin habitats. The interaction among insolation cycles is especially apparent in the paleoenvironmental records of the East African Rift System, where the longest records of human evolution are preserved. However, environmental indicators such as deep-sea oxygen isotopes, terrestrial dust flux, paleosol carbon isotopes, and lake sediments do not point consistently to any simple trend or climate driver of evolutionary change. Comparison of environmental indicators cautions against an exclusive focus on any given end-member of environmental fluctuation (driest or wettest, warmest or coolest), and argues for the impact of the entire range of variability in shaping evolutionary change. A model of alternating high and low climate variability for tropical Africa further implies that specific environmental indicators reflect different aspects of East African environmental dynamics. The model may thus help reconcile some of the conflicting interpretations about the environmental drivers of hominin evolution. First and last appearances of hominin lineages, benchmark biogeographic events, and the emergence of key adaptations and capacities to alter the surroundings are consistently concentrated in the predicted longest intervals of high climate variability. The view that emerges is that important changes in stone technology, sociality, and other aspects of hominin behavior can now be understood as adaptive responses to heightened habitat instability.

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1. Introduction

In recent decades the relationship between environmental and evolutionary change has become prominent in the thinking of Earth scientists and biologists interested in human evolution. As researchers have investigated this relationship, the goal has largely focused on whether evolutionary events have corresponded in time and space with particular types of environmental events.

Evolutionary events include morphological change, speciation (i.e., the division of a lineage into two distinct species), lineage extinction, and the origin of novel behaviors and ecological interactions. Environmental events include directional shifts in habitat (e.g., drying, cooling) and also increases or decreases in environmental variability. Because adaptive evolution, lineage divergence, and extinction are the results of evolutionary processes that play out over extended periods of time, the co-occurrence between environmental and evolutionary events can be expected to take place within intervals spanning 10^3–10^5 years rather than in exact 'instants' in time.

Although the analysis of temporal associations is helpful, the ultimate goal of this research is to understand the processes by which environment may have affected the major features of human evolutionary history. The co-occurrence between environmental and evolutionary events is, by itself, insufficient in building this understanding. It is essential to point explicitly to how environmental change led to particular genetic, demographic, and other mechanisms that drive evolutionary responses. That human evolution has largely been unaffected by the external environment also needs to be considered, an idea sometimes referred to as the evolution-environment 'null hypothesis' since it implies that evolutionary change can, and has, taken place at various times independent of environmental change (National Research Council, 2010; Potts, 2012a). The hypothesis that environmental events have indeed influenced evolutionary change is, nonetheless, consistent with natural selection as the process that has shaped the adaptations of organisms, including hominins. It is also consistent with habitat expansion, fragmentation, and disappearance as factors involved in the origination, divergence, and extinction of populations.

Although it is sometimes said that Darwin established a link between climate and evolution, his scenario of human origins

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(Darwin, 1871) was devoid of the idea that environmental change directly stimulated the variety of evolutionary events that happened over the course of human evolution. Throughout much of the 20th century, paleoanthropologists followed his lead. Rather than focusing on external environmental stimuli, researchers favored intrinsic explanations of human evolution (e.g., Bartholomew and Birdsell, 1953; Washburn, 1960), in which a single environmental transition from forest to grassland set the stage for the emergence of bipedal walking and tool use. This earliest shift in way of life then laid the foundation for meat-eating, hunting, brain expansion, control of fire, food sharing, complex social interactions, and language, among other novel adaptations that characterize how human ancestors interacted with the world. Such explanations have generally looked no further than prior adaptations to explain, through complex feedback, how evolutionary developments early in time could account for the evolution of later ones as ancestral hunter-gatherers gradually overcame the challenges of the African savanna and Eurasian ice ages (e.g., Klein, 1989; Wilson, 1978; Wolpoff, 1980; see Potts, 2012a for further discussion).

A heightened interest in extrinsic explanations, by contrast, developed in the 1980s (Laporte and Zihlman, 1983), especially through the writings of Elisabeth Vrba (1985a,b, 1988, 1992, 1995a,b; Vrba et al., 1989, 1995). Such explanations focus on factors external to the organism in explaining how evolutionary change was initiated, and in accounting for the tempo and mode of evolution, the turnover of species (originations and extinctions), and the specific adaptations that emerged in particular intervals of time.

The most prominent statements that invoke extrinsic causes of human evolutionary change are the turnover-pulse hypothesis of Vrba (see references above), the variability selection hypothesis proposed by Potts (1996a,b, 1998a,b, 2007, 2012a,b), the aridity hypothesis developed by deMenocal (1995, 2004, 2011) and deMenocal and Bloemendal (1995), and the deep-lakes hypothesis of Trauth and colleagues (Trauth et al., 2005, 2007, 2010). A variety of other studies published in recent years offer variations on the themes established by these four main hypotheses and identify factors responsible for creating spatial and temporal variability in habitats and fossil assemblages (e.g., Bailey et al., 2011; Kingston, 2007; Cerling et al., 2011; Hopley and Maslin, 2010; Passey et al., 2010).

This suite of hypotheses is obviously Africa-oriented and heavily weighted toward East Africa. Two reasons why Africa-based researchers have been so active in proposing environment-evolutionary linkages are: (1) the relatively precise geochronological resolution of lengthy stratigraphic sequences, especially in East Africa, which preserve hominins, other fossil organisms, and multiple paleoenvironmental indicators; and (2) the totality of paleontological and archeological evidence, which strongly implies that Africa was the source of many of the critical transitions in human evolution, including the origin of the hominin clade; each genus within the clade; the extant human species; and key adaptations including bipedality, stone technology, and several other behavioral innovations.

The purpose of this article is to review the data sets that have led to the most prominent ideas linking environmental and evolutionary change. It also identifies the areas of agreement and disagreement among these ideas and data interpretations. A framework for tropical African climate variability is proposed as a way of reconciling the diverse data sets; it thus has promise as an analytical tool for evaluating environment-evolutionary linkages, particularly relevant to East African hominins and contemporaneous organisms. This predictive framework describes how intervals of high and low climate variability alternated over the past several million years, and provides insights into the adaptive history of African hominins, including the emergence of *Australopithecus*, *Homo*, and *Homo sapiens*.

2. The problem of explaining one-time evolutionary outcomes

Since the purpose here is to review environmentally-based explanations of human evolution, attention must be given to a simple fact of our evolutionary history that creates an apparently insurmountable challenge in searching for testable explanations of any kind. Hominin evolution has yielded only a single surviving species, *H. sapiens*. If the point of paleoanthropology is solely to explain how *H. sapiens* emerged, human evolution becomes a singular, hard-to-comprehend phenomenon, a unique event largely intractable to scientific analysis. The flaw in this characterization is that almost all evolutionary phenomena can be narrowly construed as one-time outcomes. The evolution of any lineage could be defined as a singular phenomenon, thus creating a scientific predicament since scientific explanations require the analysis of phenomena that are, in some sense, repeatable and capable of examination as multiple instances for which explanatory factors or variables can be explored and tested. However, the evolutionary events that define any given clade, including hominins, are comprised of recurrent instances of adaptive change, multiple episodes of speciation and extinction, and a series of biogeographic events (e.g., dispersals, range expansions and contractions) that can be examined in relation to various types of environmental change or shifts in climate mode.

Thus the problem with the claim that hominin evolution represents a one-time evolutionary outcome is that it ignores the multitude of events that have occurred in the hominin clade and in the ancestry of *H. sapiens*. It is this series of events, or natural experiments, which enables researchers to assess multiple instances and patterns of evolutionary change and to test for explanatory variables. Table 1 offers a synopsis of several of the key adaptive and phylogenetic events in hominin evolution. The timing of speciation and extinction events and the onset (i.e., the increase and spread) of novel adaptations can be estimated by first appearance datums (FADs) and last appearance datums (LADs), indicated by the first or last in a particular sample of fossils that has morphological similarity (thus defining a lineage distinct from others) or by archeological indicators that denote a novel behavior or ecological innovation.

The overarching picture of hominin evolution entails the emergence of a small-bodied bipedal ape, probably in tropical/subtropical latitudes of Africa, followed by 6 million years of multiple episodes of species diversification and extinction; the evolution of adaptations that temporarily improved how species of *Australopithecus*, *Paranthropus*, and *Homo* interacted with their surroundings; and, ultimately, the survival of a single African species that has become cosmopolitan and capable of pervasive modification of its surroundings. Any explanatory model of hominin evolution that includes the physical and biotic environment needs to account for the evolutionary pressures underlying the adaptive properties of the diverse hominin species, the suite of extinctions, and, ultimately, the survival of a species that practices an astonishing degree of niche construction via cultural buffering of its environment (Laland et al., 2001; Odling-Smee et al., 2003; Potts, 1996a, 2012a).

The fact that hominin evolution has entailed numerous phylogenetic and adaptive events (Tables 1 and 2) does not mean that paleoanthropologists are close to having a sufficiently rich data set of events that is well-resolved in time and space and which might then lead to a straightforward understanding of human evolution. The pace and nature of paleoanthropological discoveries denote

Table 1
Key adaptive and biogeographic events in hominin evolution, and their approximate timing based on current fossil and archeological samples. FADs and LADs of species are shown in Table 2. Ma: million years ago; ka: thousands of years ago.

Evidence of the prevalence or spread of novel behaviors	
Adaptive events	
Morphologies favoring bipedal walking coupled with tree-climbing	7.0–3.4 Ma
Morphological specializations favoring obligate bipedality	4.2–1.5 Ma
Oldest careful stone flaking coupled with transporting of stone and food (Oldowan technology)	By 2.6 Ma
Use of tools to increase protein/fat acquisition from large animals	By 2.6 Ma
Expansion in stone transport behavior (>10 km)	2.0 Ma
Initial expansion in cranial capacity in the genus Homo	2.0–1.5 Ma
Increased mobility; elongated legs and taller stature	2.0–1.7 Ma
Occupation of novel ecosystems in temperate latitude Asia and tropical Southeast Asia	1.9–1.7 Ma
Origin of large cutting tools (Acheulean technology)	By 1.76 Ma
Manipulation and control of fire, including oldest hearths	1.0–0.8 Ma
Most rapid expansion in cranial capacity; slower dental development; prolonged life history	1.0–0.2 Ma
Early ground shelter-building	By 400 ka
Increased sophistication of toolkit/technology (Middle Stone Age technology)	300–200 ka
Increased cultural differentiation	200–20 ka
Increased innovation in hunting and other foraging activities, including fishing tools	100–50 ka
First obvious symbolic expression using preservable materials	100–50 ka
Cultivation of plants; domestication of animals; first human-dominated ecosystems	18–10 ka
Biogeographic events	
Within-Africa dispersal of late Miocene bipeds	7.0–5.8 Ma (diversification of *Sahelanthropus, Ardipithecus*, and *Orrorin* from eastern to north-central Africa)
Spread of Australopithecus within sub-Saharan Africa	3.5–2.8 Ma
Initial spread of *Homo* within sub-Saharan Africa	2.6–1.9 Ma
Spread of Oldowan behavior from eastern to southern and northern Africa	~2.0 Ma
Spread of *Homo* to Eurasia	1.9–1.7 Ma (initial excursion of *Homo* sp., possibly *H. erectus*, and basic core/flake technology beyond Africa across western to eastern Asia) 1.2 Ma (initial excursion of large cutting tools into Asia) 0.5 Ma (initial excursion of large cutting tools into NW Europe)
Possible back-migrations of *Homo* to Africa	1.5 Ma (possible re-introduction of *Homo erectus* morphology to Africa from Asia)
Spread of *Homo sapiens* beyond Africa	135–100 ka: initial excursion of populations across western to eastern Asia 80–60 ka: beginning of principal worldwide expansion of populations from Africa

a sharply-inclined learning curve. It is particularly important to note that the evolutionary events listed in Tables 1 and 2 have shifting age constraints, although with increasing precision for some of these events. Thus the FADs, LADs, and temporal durations given Table 2 are estimates with uncertainties. An understanding of how environmental events and particular evolutionary innovations are associated is very likely to change as discoveries occur.

A final point about testing the environment–evolutionary relationship follows from the fact that the evolution of hominins, like all other organisms, has taken place in an ecological context involving other organisms. Thus, the evolutionary histories of other large-bodied mammals that co-existed with early hominins provide an enlarged perspective about how change in the physical and biotic environment may have affected human ancestors and the ecosystems in which they lived. Analysis of the vertebrate evolutionary context of hominin evolution has emerged as a vibrant research area (e.g., Behrensmeyer et al., 1997; Bobe and Behrensmeyer, 2004; Bobe et al., 2007; Potts, 1998a; Reed, 1997; Vrba, 1995b; Werdelin and Sanders, 2010). As noted below, in a number of instances, these studies have produced important insights into the selection pressures involving climate and vegetation change that co-occurred with the evolution of Plio-Pleistocene *Homo* and the origin of *H. sapiens* in the late middle Pleistocene.

3. Methods

3.1. Key data sets

Researchers have proposed a variety of environmental drivers of human evolutionary change. The principal ideas include climatic cooling, persistent warmth, pulses of aridity that included the spread of C_4 grasses, phases of high rainfall and humidity, and variability itself as causal forces in evolution. These environmental patterns imply particular trends or oscillations in hydrology, landscape, and vegetation, and it is these shifts in resources and ecological setting that are thought to initiate evolutionary change. The key data sets used in developing these hypothetical drivers of evolution are:

1. Deep-sea oxygen stable isotopes, $\delta^{18}O$ (e.g., Lisieki and Raymo, 2005; Shackleton, 1995; Zachos et al., 2001)
2. Eolian terrestrial dust records, also measured in deep-sea cores (e.g., deMenocal, 1995, 2004; deMenocal and Bloemendal, 1995)
3. Molecular plant biomarkers (e.g., Feakins et al., 2005, 2013)
4. East African paleosol stable carbon isotopes, $\delta^{13}C$ (e.g., Cerling, 1992; Cerling et al., 2011; Kingston, 2007; Levin et al., 2004, 2011; Passey et al., 2010; Wynn, 2004)
5. East African lacustrine diatomites and clays (e.g., Ashley, 2007; Deino et al., 2006; Trauth et al., 2005, 2007; Owen et al., 2008)

Table 2
Hominin species originations and extinctions: First appearance datums (FADs) and last appearance datums (LADs) of hominin species, their geographic range, temporal duration, and whether current FADs or LADs for East African lineages occur in a predicted high or low climate variability interval.

Species	Geographic distribution	FAD (Ma)	LAD (Ma)	Temporal duration (kyr)	East African FAD/LAD during high or low climate variability
Homo sapiens	African origin; now cosmopolitan	0.20		200	FAD: high
H. neanderthalensis	Eurasia	0.20	0.03		na
H. floresiensis[b]	Flores (Indonesia)	0.10	0.02	85 [1000]	na
H. heidelbergensis[c]	Europe/Africa (Asia?)	0.70	0.30	400	na
H. antecessor[a,d]	Western Europe	1.20	0.80	400	na
H. erectus[e]	Africa/Asia (Europe?)	1.89	0.10	1800	FAD: high
H. habilis	Eastern-southern Africa	2.40	1.40	900	FAD: low LAD: high
H. rudolfensis[a]	Eastern Africa (Kenya-Malawi)	2.30	1.80	500	FAD: high LAD: high
Paranthropus robustus	Southern Africa	1.80	1.20	600	na
P. boisei	Eastern Africa (Ethiopia to Malawi)	2.30	1.20	1100	FAD: high LAD: high
P. aethiopicus	Eastern Africa	2.72	2.30	400	FAD: high LAD: high
Australopithecus sediba[a]	Southern Africa	1.97	1.97	?	na
Au. garhi[a]	Eastern Africa	2.50	2.50	?	High
Au. africanus	Southern Africa	2.80	2.10	700	na
Au. afarensis[f]	Eastern Africa	3.80	3.00	850	FAD: high LAD: high
Au. anamensis	Eastern Africa	4.20	3.90	300	FAD: high LAD: high
Ardipithecus ramidus	Eastern Africa	4.40	4.2–4.4	200	FAD: low LAD: high
Ar. kadabba[a]	Eastern Africa	5.8–5.2	5.8–5.2	?	?
Orrorin tugenensis[a]	Eastern Africa	6.2–5.8	6.2–5.8	?	?
Sahelanthropus tchadensis[a]	North-central Africa	7.0–6.0	7.0–6.0	?	?

na: not applicable, non-East African species.
[a] Poorly documented morphological variation and distinction from other taxa, and/or poorly documented temporal and geographic range, especially for taxa known from only a single fossil site; several other taxonomic names not listed here also have these limitations.
[b] FAD based on fossils is ~95 ka, although a toolmaking species on Flores reaches back to ~1.02 Ma; revised LAD is under study.
[c] Includes *H. rhodesiensis*, sometimes viewed as the African branch of the species that gave rise to *H. sapiens*; earliest members may include *H. antecessor*, dated >0.8 Ma in western Europe.
[d] May represent early populations of *H. heidelbergensis*.
[e] Includes early African *H. erectus*, sometimes referred to as *H. ergaster*, and Georgian (Dmanisi) hominins sometimes separated out as *H. georgicus*.
[f] Species diversity in the time period of *Au. afarensis* is poorly known, with *Kenyanthropus platyops* and a fossil foot indicative of locomotor diversity from Woranso-Mille, Ethiopia, suggestive of multiple lineages.

6. Eastern Mediterranean sapropels, which reflect Nile flooding and monsoonal dynamics encompassing the northeast quadrant of the African continent (e.g., deMenocal, 2004; Emeis et al., 2000; Hilgen et al., 1999; Lourens et al., 1996; Rossignol-Strick, 1983).

Fossil pollen (e.g., Bonnefille, 1995, 2010; Bonnefille et al., 2004), sediment geochemistry (e.g., Owen et al., 2011), and strontium isotopes (e.g., Joordens et al., 2011), among other types of records from fossil hominin sites, have also proved valuable in interpreting African paleoenvironments. These records, however, are typically discontinuous and offer less complete sequences of environmental dynamics over time than the six indicators noted above. The analysis here relies on comparing these six indicators, explores the possibility of integrating them into a robust model of tropical and subtropical climate dynamics, and examines how the major features of African hominin evolution relate to this model. Detailed methods of stable isotopic, terrigenous dust, biomarker, diatom, and sapropel analysis are provided in the references cited above.

3.2. Olorgesailie: a lengthy mid-Pleistocene environmental record

One inspiration for the analysis presented here is the long-term environmental and paleoanthropological research conducted in the Pleistocene Olorgesailie basin, southern Kenya Rift Valley. The strata of the Olorgesailie Formation preserve a lengthy record of Acheulean stone technology, typified by large cutting tools such as handaxes and associated simple flake/core tools; fossil mammal remains indicative of ecological transitions during the mid-Pleistocene; and paleoenvironmental indicators, including diatoms and the lithologies spanning from ~1.2 to 0.5 Ma (Behrensmeyer et al., 2002; Deino and Potts, 1990; Isaac, 1977; Owen et al., 2008; Potts, 1989, 1994, 2007; Potts et al., 1999, 2004). In a recently-recognized upper sedimentary formation, Acheulean technology is replaced by a smaller, more diverse toolkit of the early Middle Stone Age, and the early Pleistocene mammalian fauna is supplanted by a modernized fauna devoid of large-bodied, specialized grazers (Behrensmeyer et al., 2002; Brooks et al., 2007; Faith et al., 2012; Potts, 1998a).

Sediments in the 300-km^2 Olorgesailie basin are sufficiently exposed to allow recognition of widespread transitions in basin hydrology and sedimentary regime. These transitions are observable on scales of several kilometers of outcrop, and are considered to reflect basin-wide and possibly regional-scale reorganizations of the environment due to a combination of climate change, tectonic deformation, and volcanic activity, with intervening periods of landscape stabilization (Behrensmeyer et al., 2002; Potts, 2001). These environmental transitions are identified by shifts between lake- and fluvial-dominated sedimentary regimes; changes between erosional, depositional, and soil-forming environments; and episodes of thick tephra deposition (≥1 m of ash or pumice). The stratigraphic positions of these 'first order changes' within the Olorgesailie Formation have been well studied (Fig. 1). The concentration of these transitions is employed as a test case concerning the model of tropical African climate dynamics considered here.

4. Results

4.1. Cooling, warmth, aridity, moisture, and variability: drivers of human evolution?

4.1.1. Cooling

The record of oxygen stable isotopes shows an overall enrichment in $\delta^{18}O$ in benthic marine foraminifera over the past 6 million years (e.g., Lisieki and Raymo, 2005, 2007; Meyers and Hinnov, 2010; Shackleton, 1995; Zachos et al., 2001). This $\delta^{18}O$ evidence of ocean

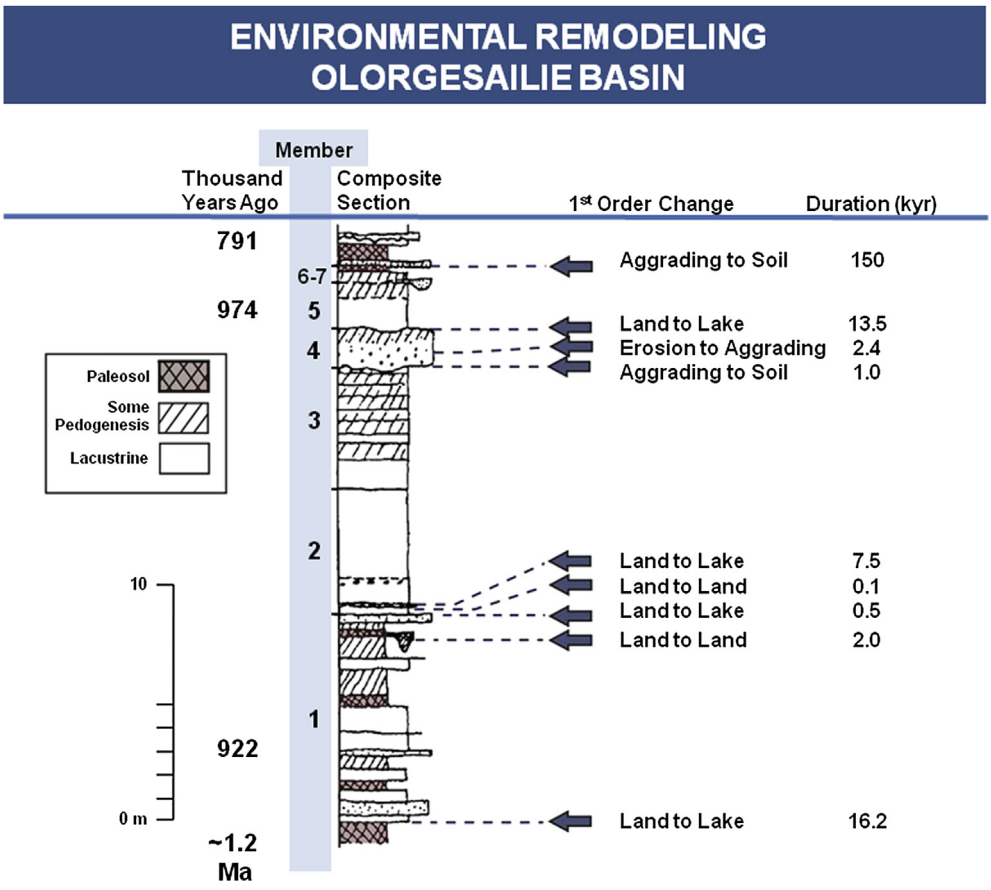


Fig. 1. Composite section of Olorgesailie Formation Members 1 through 8, dated by $^{40}Ar/^{39}Ar$ analysis and magnetostratigraphy to ~1.2–0.78 Ma (Behrensmeyer et al., 2002; Deino and Potts, 1990; Tauxe et al., 1992). Arrows indicate the stratigraphic boundaries of major reorganizations of the landscape and hydrology, called first order changes. Nine of the 21 first order transitions identified in this stratigraphic interval are depicted. The estimated duration of each environmental state is given (right side) (Potts, 2001).

cooling, intensification of Northern Hemisphere glaciation, and periodic sequestering of ocean water on land in the form of glacial ice denote a long-term trend coincident with many of the major developments in hominin evolution. Vrba's initial formulation of the turnover-pulse hypothesis focused on global cooling around 2.5 Ma as the major forcing of habitat change in Africa associated with the emergence of *Homo*, *Paranthropus*, stone toolmaking, and significant losses and gains in bovid and rodent species (Vrba, 1985a,b; 1988). Later developments of the idea (a) widened the temporal window (2.8–2.4 Ma) for the proposed turnover pulse in bovids, rodents, and hominins; and (b) added the concept of 'habitat theory', which specified the timing of extinction of warm/moist-adapted species coupled with speciation favoring cool/arid-adapted taxa (e.g., Prentice and Denton, 1988; Vrba, 1988, 1992, 1995a).

In the turnover-pulse hypothesis, the key mechanisms linking climate and evolution are directional selection, which promoted adaptations to cool grasslands, and the repeated fragmentation and coalescence of savanna habitats, which stimulated population vicariance, reproductive isolation, and speciation in grazing mammals and hominins. Vrba (1994) also speculated that cooling produced a developmental effect (neoteny) whereby immature neurocranial form was retained, favoring an evolutionary increase in brain size in *Homo*.

4.1.2. Warmth

A different perspective stems from a study by Passey et al. (2010) of paleosol carbonates in the Turkana basin of southern Ethiopia and northern Kenya. Employing a method of clumped isotopes, these investigators provide evidence that surprisingly warm soil temperature of >30 °C, and often 35–40 °C, persisted over the past 4 million years. They argue that paleosol and carbonate formation is evident throughout the Turkana stratigraphic record, and that soil temperature closely tracks mean annual temperature. Thus temperatures as warm as, or even warmer than, present-day Turkana (which is among the hottest 1% of continental locations on Earth) characterized the long era of evolving hominin populations in this region. Accordingly, persistent warmth rather than cooling was a pervasive adaptive challenge in hominin evolution from the early Pliocene onward.

These authors imply that such continual warmth led to habitat-specific, directional selection promoting the evolution of unusual aspects of human thermoregulation, such as upright, linear bodies coupled with exposed skin and sweating capacity. A genetic investigation by Rogers et al. (2004), however, points to an age constraint close to 1.2 Ma for the evolution of one of the determinants (the MC1R gene) of dark skin pigmentation in all humans, a feature expected to be under strong positive selection as soon as relatively bare skin was exposed to hot, open conditions.

4.1.3. Aridity

A long-term trend toward aridity and open vegetation has dominated thinking about Pliocene through Pleistocene environments of Africa for many decades. The case for aridity as the most important environmental factor in hominin evolution arises from

Vrba's analysis linking global cooling with African aridification. In recent years, this position has received strongest support from the analyses of wind-blown continental dust preserved in ODP Sites 721/722 in the Arabian Sea and DSDP Site 231 in the Gulf of Aden (deMenocal, 1995, 2004). Large seasonality in rainfall promotes the eolian transport of mineral dust from the African continent to adjacent oceans. deMenocal's influential analysis of the terrigenous input at East African deep-sea drill sites indicates substantial increases in dust production in three major steps: 3.0–2.6 Ma, 1.8–1.6 Ma, and 1.2–0.8 Ma (deMenocal, 1995, 2004).

The resulting idea that Plio-Pleistocene hominins emerged as part of the arid biota of Plio-Pleistocene Africa has gained support from $\delta^{13}C$, fossil bovids, and plant-wax biomarkers. The first two periods of African aridification were associated with increases in C_4 vegetation (tropical grasses) denoting the development of East African savanna (Cerling and Hay, 1986; Cerling, 1992). Study of fossil mammals in the Turkana basin shows a net increase in grazing bovids coincident with stronger aridity after 3.0 Ma, particularly in the second interval between 1.8 and 1.6 Ma (Bobe and Behrensmeyer, 2004; Bobe and Eck, 2001). Furthermore, plant leaf wax biomarkers recovered from DSDP Site 231 indicate an overall increase in $\delta^{13}C$ after 3 Ma (Feakins et al., 2005). While this finding was originally thought to denote the expansion of arid savanna grassland after this date, later analysis by Feakins et al. (2013) suggests that plant leaf wax $\delta^{13}C$ is decoupled from grassland expansions and contractions based on grass pollen percentage (Bonnefille, 2010). Feakins et al. (2005, 2013) further note that the variance in biomarker $\delta^{13}C$ fluctuates from one time interval to another, and they suggest that these fluctuations likely correspond to insolation-driven oscillations in vegetation.

The aridity hypothesis frames evolutionary change as a response to strong directional selection, which promoted the spread of novel adaptations associated with grassland expansion, and also habitat fragmentation leading to population vicariance and speciation of arid-adapted fauna (Vrba, 1995a). According to deMenocal (1995, 2004, 2011), and in agreement with Vrba (e.g., 1988), aridity increases in East Africa coincided with hominin speciation, especially the first appearance of *Homo* and *Paranthropus*, the emergence of Oldowan and Acheulean tool industries, and the development of meat-eating and other dietary shifts. The aridity hypothesis, therefore, places strong emphasis on the idea articulated by Reed (1997) that "*Homo* species appear the first to be adapted to open, arid environments."

4.1.4. Moisture and deep lakes

A distinctly different perspective arises from a comparative study of lacustrine sediments in East African Rift basins. The accumulation of diatomaceous sediments in these basins indicates the periodic expansion and deepening of lakes due to monsoon precipitation (Deino et al., 2006; Kingston et al., 2007; Trauth et al., 2005, 2007, 2010). According to the initial data interpretation presented by Trauth et al. (2005), key events in hominin evolution took place during three major humid periods when deep lakes occurred as either persistent features of the rift landscape or periodic features driven by precessional forcing. These deep, expansive lakes occurred at 2.7–2.5 Ma, 1.9–1.7 Ma, and 1.1–0.9 Ma.

The timing of these moisture intervals overlap with deMenocal's arid (high dust) phases. However, in a statistical reanalysis of the dust flux data from ODP Sites 721/722 and other deep-sea records, Trauth et al. (2009) conclude that there is only a very weak positive trend toward higher dust values over the past 4.5 myr, and that strong dust increases and aridification cannot be demonstrated at 3.0–2.6 Ma and 1.8–1.6 Ma. Thus the critical evolutionary events that deMenocal ascribes to aridity are instead attributed by Trauth et al. (2007, 2010) to strong moisture-aridity variability that included expanded East African lakes. The precessionally-driven formation of large lakes is considered to have caused the periodic expansion and division of hominin populations, which elicited speciation, dispersal, and adaptive change, including dietary shifts related to brain expansion. Although wet phases and large lakes have been the focus of the work by Trauth and colleagues, their emphasis on heightened wet–dry variability is in accord with variability selection as a driving factor in hominin evolution (Trauth et al., 2010; Donges et al., 2011).

4.1.5. Variability

The variability selection hypothesis states that the adaptations of hominins and other organisms have been shaped by the instability of environmental and survival conditions over time and space (Potts, 1996a,b, 1998a,b, 2007, 2012a,b). It posits how populations (and their gene pools) may evolve during periods of greatly increased environmental variability: As the instability of Darwinian fitness escalates in a particular span of time, adaptable phenotypes (and their underlying genetic basis) are favored over habitat-specific adaptations. The process of variability selection thus enables the evolution of organisms better at adjusting to environmental variability through their physiological, behavioral, and ecological plasticity (Potts, 1996b, 1998a, 2002). This hypothesis thus seeks to explain how organisms may increase their adaptive versatility in response to the full range of environmental dynamics, and evolve the capacity to adjust to novel habitats that arise as the environment varies or populations disperse to new places.

With regard to human evolution, the implication is that early hominin populations experienced shifting conditions of natural selection and, in some lineages, certain combinations of genes enhanced the ability of those gene pools to thrive as habitats changed. The behaviors and ecological characteristics of these populations favored their persistence over others that exhibited habitat-specific adaptations, which conferred a more limited degree of response to novel environments.

The empirical support for this idea comes from an integration of paleoenvironmental data from hominin sites, especially from long stratigraphic sequences, and from regional and global records. Evidence of strong monsoonal oscillations and their effect on East African landscapes inhabited by hominins has been found at Hadar, Olduvai, Turkana, and Olorgsailie, among other basins (Ashley, 2007; Bonnefille et al., 2004; Campisano and Feibel, 2007; Lepre et al., 2007; Owen et al., 2008; Quinn et al., 2007). On a global scale, although benthic $\delta^{18}O$ shows a long-term trend toward ocean cooling and continental glaciation, the oxygen isotope record also shows a 4-fold rise in the amplitude of oscillation over the past 6 million years, and a 3-fold increase since 3 Ma.

For the most part, data sets used to promote the cooling, warmth, aridity, or moisture explanations of evolutionary change also point to monsoonal and other factors driving environmental dynamics. The northeast African continental dust record, for example, shows evidence of alternating intervals of high and low climate variability (shifts in the amplitude of arid-moist oscillation) over time scales of 10^4–10^5 years (deMenocal, 2004). In Gulf of Aden marine sediments (DSDP 231), Feakins et al. (2013) found that the range of high-resolution plant-wax $\delta^{13}C$ variability nearly equals or exceeds several long-shifts toward C_4 plants since the late Miocene, suggesting that approximately precessional-scale variability in C_3/C_4 plants was surprisingly high in certain periods over the past several million years.

Extensive East African lakes also occurred in times of large precessional-scale variability. Maslin, Trauth, and colleagues have sought to extend the variability selection idea by focusing on three pulses of extreme climate variability linked to major transitions in the global climate system at 2.7–2.5 Ma, 1.8–1.6 Ma, and

1.0–0.7 Ma (Maslin and Christensen, 2007; Maslin and Trauth, 2009; Trauth et al., 2007, 2009). In East Africa, these global transitions are manifested as ~200-kyr-long windows of extreme wet–dry cycles during which large, deep lakes developed. This 3-pulse concept is entirely consistent with the variability selection concept of selection and evolutionary change; yet it also gives special significance to humid phases and large lakes during global transitions when strong climate variability provided a catalyst to hominin evolution (Maslin and Christensen, 2007; Maslin and Trauth, 2009; Trauth et al., 2007).

Subsequent analysis of the terrigenous dust record by Donges et al. (2011) presents slightly revised ages for the major transitions in global climate in this 3-pulse model: 3.35–3.15 Ma, 2.25–1.6 Ma, and 1.1–0.7 Ma. Although the study posits the close association of important evolutionary events with climate shifts at these times, paleoanthropologists would question whether evidence is particularly strong for any major event in the first and last of these intervals in Africa.

Although grassland expansion in East Africa is usually characterized as progressive and long-term since at least the Pliocene (e.g., Cerling, 1992; deMenocal, 1995, 2004), a synthesis of paleosol $\delta^{13}C$ records across East Africa suggests that a broad range of vegetation variability occurred over the past several million years (Feakins et al., 2013; Kingston, 2007). Using an innovative method to reconstruct the dynamics of woody cover, Cerling et al. (2011) revives the savanna hypothesis, which invokes the selection pressures of the savanna mosaic as a critical influence on human evolution. Nonetheless, these authors note that the technical definition of savanna includes habitats that exhibit from 5 to 80% woody cover. Although the term 'savanna' can thus be reconciled with other evidence of environmental variability, this revised savanna hypothesis does not clearly distinguish between habitat-specific selection (e.g., in a stable spatially-heterogeneous setting) and variability-driven processes of evolution.

Further support for the variability selection hypothesis comes from analyses by Groves (2011a,b) employing a variety of tests from theoretical biology. This work shows the concordance of variability selection with the long-term rise in amplitude evident in the benthic $\delta^{18}O$ record, and identifies 2.5–1.2 Ma as the most critical phase in hominin evolution favoring versatile genetic adaptations. Rather than a progressive amplitude increase, however, East African climate dynamics manifested many escalations in wet–dry variability interrupted by more stable climate. As discussed below, the timing of the high variability episodes sets up a series of new tests regarding the evolution of adaptability during human evolution.

4.2. Climate dynamics: a synthesis relevant to East African hominin evolution

The data summarized here raise the question about how best to reconcile the diverse interpretations of environmental influences on human evolution. The modulation of orbital precession by eccentricity has intrigued most investigators of African paleoclimate, and offers a way of resolving the apparent incompatibility of data interpretations based on different proxies. As illustrated by Fig. 2, the interaction of eccentricity and precession curves highlights a pattern of alternating high and low climate variability over time, in line with observations made by deMenocal (2004), Trauth et al. (2007, 2010), Maslin and Christensen (2007), and Potts (2007, 2012b), among other researchers. This alternating pattern reflects heightened and dampened tropical monsoon variability. Boundaries between high and low variability can be assigned on the basis of eccentricity (e), with low variability defined by $e \leq 0.0145$, i.e., 1σ below mean e for the past 5 million years. High arid-moist variability is predicted for any time interval when e is above this level

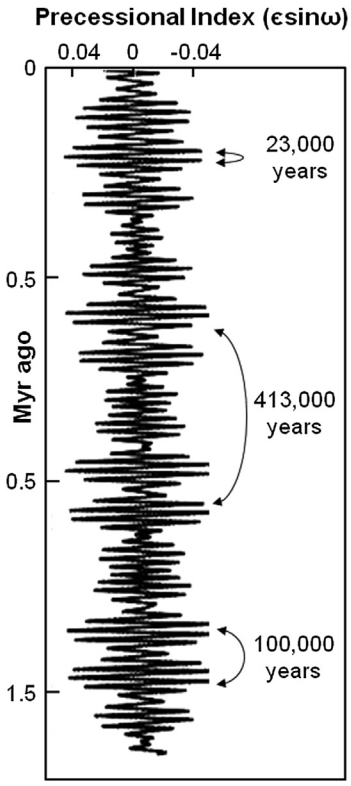


Fig. 2. The precessional index is determined by the interaction of the 23-kyr periodicity of orbital precession, and the modulation of its amplitude by eccentricity periods of ~100 and 413 kyr. The result is an alternating pattern of high and low amplitude. Eccentricity-modulated precession thus can be used to define a predictive model of high- and low-climate variability, as discussed in the text.

(Table 3). deMenocal and the author have tested this way of distinguishing high and low variability by comparing terrigenous measurements from ODP Site 721, and we find that dust flux standard deviation becomes larger or smaller in agreement with the predicted shift between high and low variability for ~76% of the pairs of adjacent intervals ($N = 66$ transitions) over the past 5 million years ($\chi^2 = 17.5$, 1 df, $p < 0.001$; i.e., the direction of change in dust standard deviation between intervals follows the predicted shift; e.g., see Table 3). Although the e value used here to define high vs. low periods tends to extend the length of high variability intervals, the resulting time boundaries are supported by empirical data, including dust flux, sapropel spectral reflectance, and East

Table 3
High/low-climate variability model for ~1.3–0.8 Ma, based on eccentricity-modulated precession, where low variability (red) is defined as $e < 0.0145$. The direction of change in standard deviation in East African ODP Site 721 dust data is used to test the predicted shifts, and show agreement for five of six transitions. Note that the predicted low variability interval from 0.902 to 0.916 Ma maintains a high σ, which suggests that ~1.29–0.82 Ma was characterized by high climate variability, with a short low variability interval of 16 kyr. Bold type denotes a high variability interval >100 kyr.

Predicted climate variability	Time interval (Ma)	Eccentricity (e)	Site 721 dust flux σ	Site 721 data sample	Agreement
High	0.824–0.900	0.0298	8.3	50	Yes
Low	0.902–0.916	0.0111	8.3	14	No
High	**0.918–1.116**	**0.0406**	**6.7**	**143**	**Yes**
Low	1.118–1.134	0.007	1.9	15	Yes
High	**1.136–1.290**	**0.0284**	**5.9**	**95**	**Yes**
Low	1.292–1.312	0.008	2.9	14	Yes

African sedimentary records. Three of the most prolonged intervals of predicted high climate variability are ~2.79–2.47 Ma (duration: ~320 kyr), ~1.89–1.69 (~200 kyr), and ~1.12–0.92 Ma (~200 kyr). Within these three intervals, Trauth and colleagues' periods of large lakes broadly overlap with deMenocal's inferred aridity phases based on dust, paleosol $\delta^{13}C$, and the prevalence of grazing bovids.

The high/low climate variability framework, based on eccentricity-modulated precession, thus may offer a way of reconciling the interpretation of both moderate-to-strong aridity and deep lakes in the same time intervals. Eastern Mediterranean sapropels currently offer the best empirical record that captures the alternation of high and low climate variability in northeast Africa over the past several million years (deMenocal, 2004; Emeis et al., 2000; Hilgen et al., 1999; Lourens et al., 1996). Fig. 3 illustrates this observation, and emphasizes the overlap of the deep lakes posited by Trauth et al. (e.g., 2007) and aridity intervals posited by deMenocal (e.g., 2004), although statistical analysis by Trauth et al. (2009) argues that aridity increases based on terrigenous dust alone were largely confined to 1.5 Ma and after.

More specific tests of the validity of dividing East African climate into eras of high and low climate variability come from long stratigraphic sequences at early hominin sites. One example is the sedimentary sequence in the Olorgesailie basin (Behrensmeyer et al., 2002; Potts, 2001). Olorgesailie preserves a quasi-continuous stratigraphic record spanning ~1.2–0.5 Ma. As noted above, deMenocal (1995, 2004) identifies 1.2–0.8 Ma as a time of marked aridification, whereas Trauth et al. (2005, 2007) defines a ~200-kyr wet phase over a more restricted period between 1.1 and 0.9 Ma. In this more restricted interval of overlap, the Olorgesailie sequence records 21 first order environmental events (basin-wide transitions in hydrology and landscape) out of a total of 30 such events between 1.2 and 0.5 Ma. In other words, 70% of the major environmental reorganizations occur within 29% (200 kyr) of the total time recorded in the Olorgesailie Formation. The prediction of strong environmental variability between 1.1 and 0.9 Ma is thus borne out by the concentration of environmental perturbations in the Olorgesailie sequence and by the eolian dust and deep lake data. Variability over sub-millennial, millennial, and precessional scales is directly apparent for this time interval in the Olorgesailie sediments (Fig. 4; Potts et al.. 1999; Owen et al., 2008).

5. Discussion

5.1. Evolutionary change in times of strong environmental variability

From the Pliocene through Pleistocene, the major features of African hominin evolution include the first appearances of three genera – *Australopithecus*, *Paranthropus*, and *Homo* – and the emergence of three major stone tool industries – the Oldowan, Acheulean, and Middle Stone Age. The two largest biogeographic events are the Plio-Pleistocene dispersal of the genus *Homo* and the late Pleistocene dispersal of *H. sapiens* beyond Africa. It is intriguing that the currently known FAD for each of these evolutionary events coincides with four of the most prolonged periods of predicted high climate variability for tropical/subtropical eastern Africa.

Using the model outlined in the previous section (Fig. 2, Table 3), eccentricity modulation of precession defines eight intervals in which high climate variability was especially protracted, ranging from approximately 190 kyr–330 kyr in duration (Potts, 2012b). The FAD for the most widespread Pliocene genus *Australopithecus*, based on fossils of *Au. anamensis* from Kenya and Ethiopia, occurs in the ~200-kyr interval of predicted high-climate variability from ~4.36 to 4.16 Ma (Leakey et al., 1998; White et al., 2006). The minimum divergence age for *Paranthropus* (~2.7 Ma), possibly the origin of *Homo*, the oldest known Oldowan stone-tool sites, and the earliest known meat acquisition from large mammals are contained in a predicted ~330-kyr-long interval of heightened instability

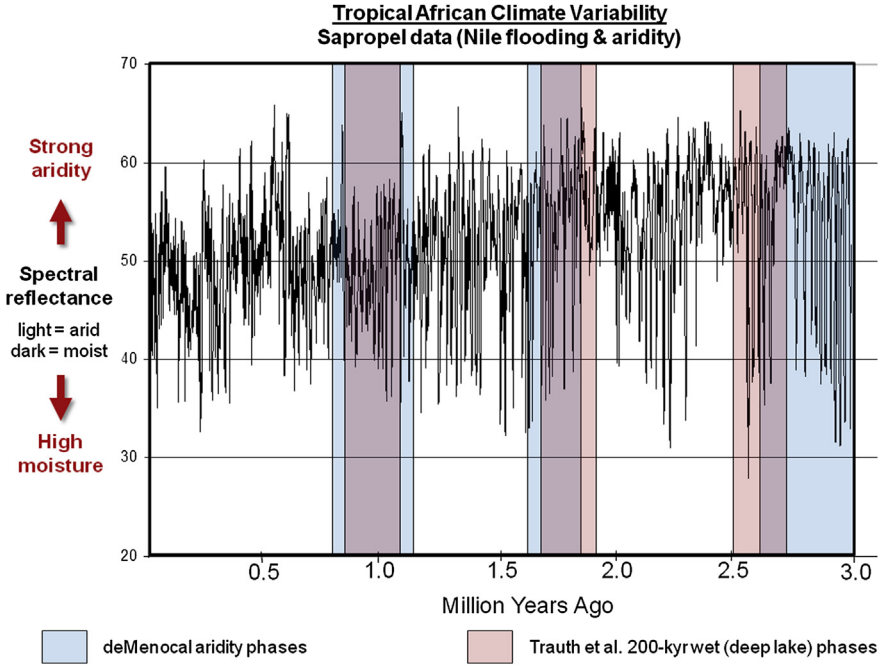


Fig. 3. Sapropel variability illustrates the presence of alternating intervals of high and low climate variability over the past 3 myr. Spectral reflectance is a measure of dark vs. light color, with higher values for lighter stratigraphic layers (arid times, prevalent during less variable periods) and lower values for darker strata (moist times; especially prevalent during high moist-arid variability). Overlapping times of strong aridity (blue; deMenocal, 1995, 2004, 2011) and high moisture (pink; Trauth et al., 2005, 2007) fall into predicted intervals of high climate variability. Spectral reflectance data from ODP Site 967A (Hilgen et al., 1999; P. deMenocal, personal communication).

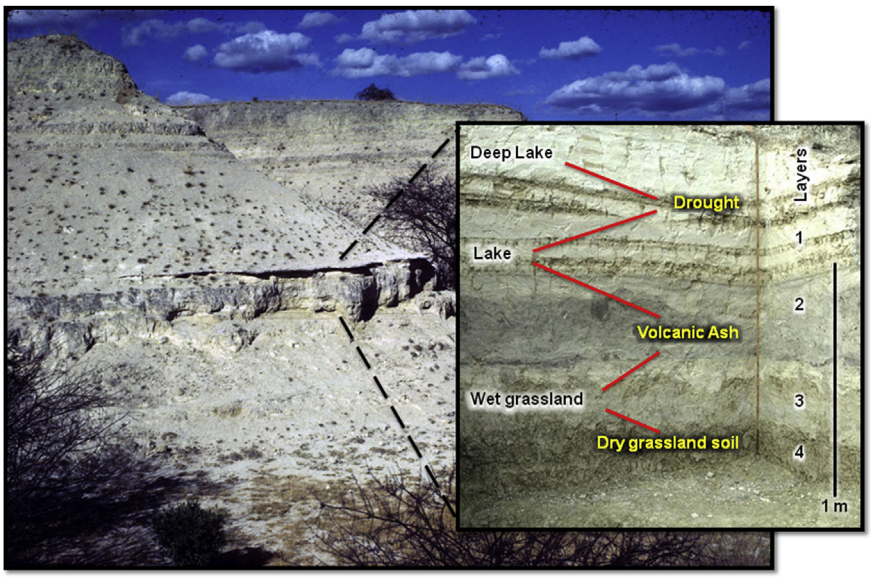


Fig. 4. One eroded hillside of the 300-km^2 Olorgesailie basin, southern Kenya Rift, representing from ~992 to 980 ka – i.e., ~12,000 years of landscape change. Inset: Lithologies, diatoms, paleosol $\delta^{13}C$, and geochemical records indicate the variability in climate and overall environment, which can be traced basin-wide within an interval ~2000 years long within this stratigraphic sequence.

between 2.79 and 2.46 Ma (Bobe and Leakey, 2009; Domínguez-Rodrigo et al., 2005; Semaw et al., 2003). The oldest Acheulean technology and possibly the oldest clear evidence of dispersal of *Homo* into Eurasia are found within the 200-kyr interval of strong wet–dry variability between ~1.89 and 1.69 Ma. Finally, the prolonged high variability interval between ~356 and 50 ka coincides with the FAD for *H. sapiens* and Middle Stone Age technology, which defines the initiation of behavioral innovation that ultimately underpins the cultural diversification of *H. sapiens* at the end of this period (Potts, 2012a).

This alternating climate variability model differs in several ways from the 3-pulse concept developed by Trauth, Maslin, and colleagues. First, the high/low variability framework defines a larger number of time windows over the course of human evolution in which significant adaptive shifts, dispersals, and species turnover events may have been stimulated by the shift to higher variability. Second, the eight most prolonged intervals of high variability are not always associated with major transitions in global climate. Last, the alternating model implies that low variability was also important; the transitions from relatively stable environment to prolonged instability may have been crucial in contesting and potentially reshaping (via variability selection) the adaptability of organisms and the resilience of their lineages.

The model of alternating high and low climate variability accommodates within the longest high variability intervals all of the broad strokes of human evolutionary history: the major branching points and geometry of hominin phylogenetic history, the main behavioral innovations in early hominin technology, and the starting points for the most significant events in the biogeographic history of *Homo*. Particularly striking is the emergence of the behaviors characteristic of *H. sapiens* along with the origination and the global dispersal of our species, all of which are currently situated in the high eccentricity interval that lasted ~306 kyr, ending ~50 ka.

This latter interval is particularly interesting in that it coincides with the revision of the large mammal biota of southern Kenya, and probably of East Africa in general (Faith et al., 2012; Potts, 1998a). This turnover entailed the loss of specialized grazers of large body mass in several mammalian groups (equids, suids, elephantids, hippopotamids, and cercopithecids), the rise of hyper-hypsodont grazing bovids, followed by their extinction at the end-Pleistocene (Faith et al., 2012; Potts and Deino, 1995). On the basis of this overall pattern of extinction, the East African grazing fauna that had dominated East Africa since ~1.8 Ma (e.g., Bobe and Leakey, 2009; Bobe et al., 2007; Reed, 1997) was reduced in diversity, first in the non-bovid and ultimately in the bovid fauna, suggesting at least a two-step process of emergent adaptability in the survivors across multiple lineages. The mammalian taxa that evolved, spread, and persisted during the high eccentricity interval were those capable of surviving recurrent large-scale environmental revisions of the middle and late Pleistocene.

In short, the repeated association of critical events in hominin evolution with the most prolonged intervals of high climate variability suggests that environmental instability had a significant role in hominin adaptive evolution and the emergence of adaptability in the genus *Homo*.

There are good reasons, though, why this point cannot yet stand as a firm conclusion but requires ongoing testing. First, on the evolutionary side, the dates of FADs are likely to change; further discoveries may shift the first appearances of species or adaptive innovations back in time such that they vacate the prolonged high variability intervals in which they now reside. Second, stratigraphic gaps, taphonomic biases, and time averaging make it impossible at present to test for exact associations between evolutionary events and specific orbital cycles, e.g., the dry vs. wet portions of precessional cycles (Behrensmeyer and Hook, 1992; Hopley and Maslin, 2010; Vrba, 1995a). Correlation between evolutionary and environmental events can be investigated in only a limited way given the poor resolution of the appearances and disappearances of species lineages and behaviors in the fossil record. At the same time, evolutionary change takes place over time intervals that may be brief or lengthy. The variability selection model of adaptive change, for example, posits that assembling the genetic variations responsible for more versatile adaptations, followed by the increase and spread of adaptable organisms, can take place over sustained spans of heightened environmental instability. The process of evolving and expressing this type of adaptive change is thus not expected to have occurred in its entirety in any one precessional cycle.

Third, on the environmental side, it is essential to test whether the climate signals predicted by the high/low variability model are truly recorded as variable landscapes in the basins and strata where evidence of hominin populations is found. The example discussed earlier concerning Olorgesailie illustrates that this point can be tested by the temporal distribution of basin-wide landscape changes. However, in that particular example of intense landscape remodeling between 1.1 and 0.9 Ma, evidence for any major event in East African human evolution is not particularly strong. Further work may show that this period coincided with the LAD for *Paranthropus* and/or the dispersal of the Acheulean out of Africa. Both of these events currently reside ~1.2 Ma, near the outset of this long high variability interval, and coincide with the first of two lengthy intervals of predicted climate instability (Table 3). Within the Olorgesailie basin, furthermore, research has documented a shift in how hominin toolmakers used the landscape between 1.1 and 0.9 Ma (Potts et al., 1999).

Although for these reasons it is not yet certain whether strong environmental instability typically caused evolutionary change to occur, the high/low variability model implies that hominin populations repeatedly confronted lengthy periods of shifting hydrology, associated with the episodic development and disappearance of lakes/wetlands, soils, and conditions of floodplain aggradation and fluvial erosion. This instability connotes highly irregular landscapes and patterns of food abundance and distribution over time, which likely triggered marked shifts in population density and inconsistencies in Darwinian selection. The idea underlying variability selection – that organisms, including hominins, can respond to highly variable environments by evolving ways of enhancing their adaptive versatility – is consistent with this high/low variability model.

5.2. The pattern of species extinctions

The diversity of hominin species and the occurrence of simultaneous lineages are reminders of the important role of extinction in shaping hominin phylogenetic history (Table 2). If we consider the emergence of stone technology, an enlarged brain, food-sharing behavior, innovations such as fire control and cooking, and complex sociality, there is no doubt about the overall widening of adaptability over time in the evolution of *Homo*. However, the reduction of lineage diversity to a single species suggests that the adaptability of hominin species had its limits.

Table 2 summarizes what is currently known about the first and last appearances of taxa and, for African species, whether these FADs and LADs are temporally associated with predicted high or low climate variability intervals. The records of African fossil hominin lineages tend to begin and end in periods of high climate variability. Four examples illustrate, moreover, the role of adaptability and behavioral options in deciding the fate of contemporaneous lineages. First, stone toolmakers, including early members of the genus *Homo*, outlasted lineages that did not apparently make stone tools on a regular basis and thus apparently lacked access to the expanded array of foods made available by sharp and effective crushing implements. Second, the expanded neurological capabilities of *Homo erectus* outlasted the massive chewing apparatus of its contemporary *Paranthropus boisei*. Although the enlarged jaw and massive chewing muscles of *P. boisei* could consume a wide diet, brains provide a far more powerful means of adjusting to environmental stimuli and novel challenges in any mammal (Navarrete et al., 2011). Third, looking outside of Africa, the ultimate extinction of *H. erectus* and *Homo floresiensis* took place on small islands, where the options for overcoming resource uncertainty are known to be limited.

The extinction of *Homo neanderthalensis* in Eurasia provides a fourth and final example. Despite recent evidence for slight genetic admixture with *H. sapiens*, the extinction of all Neanderthal populations was a real event that took place by ~30 ka (e.g., Finlayson et al., 2006; Green et al., 2010; Mellars, 2004). Neanderthals had physical adaptations that enabled them to exploit the cool, wooded environments of Europe and, on occasion, the Near East. However, strong oscillations accompanied environmental deterioration toward the end of OIS-3 in Europe, from ~50 to 24 ka (Guiot et al., 1989; van Andel, 2003). As European Neanderthals remained south of the advancing cold landscapes, populations of *H. sapiens* entered these harsh habitats. As they did so, *H. sapiens* in other parts of the world also invaded rain forests and deserts – i.e., far wetter, drier, and warmer landscapes than the Neanderthals or any previous hominin species could inhabit. *H. sapiens* from Africa had evolved the capacity to alter and adjust to its local settings in unprecedented ways through cultural buffering and behavioral diversification.

Given the prolonged interval of high-climate variability in Africa that immediately preceded the start of the worldwide dispersal of *H. sapiens*, including dramatic wet–dry oscillations of so-called African 'mega-droughts' (Cohen et al., 2007; Scholz et al., 2007), it is tempting to posit that strong environmental variability refined the neurological and social underpinnings of the cultural capabilities of *H. sapiens*, including an effectiveness in modifying the immediate habitat as a baseline survival mechanism. This manner of interacting with the surroundings ultimately succeeded in colonizing habitats worldwide. Thus, even at a pre-industrial level, the dispersal of *H. sapiens* sowed the seeds for human global impact.

6. Conclusion

Understanding the environmental drivers of human evolution hinges upon evolutionary processes that are consistent with empirical environmental data. The development of paleoenvironmental records is important because paleoclimate, vegetation, and faunal and landscape change are likely to reflect, and potentially predict, the selective forces that shaped the survival and extinction of early hominin populations over time and space.

Comparison of a several key records suggests that some (e.g., paleosol $\delta^{13}C$, percent grazing bovids, eolian dust) can be effective in sensing an aridity signal whereas others (e.g., lake sediments) focus on moisture. By themselves, different indicators have been used to advocate either cool, warm, arid, or humid environments as crucial drivers of evolutionary change. By combining the indicator records into a framework of environmental dynamics, the complexly nested variability of evolutionary conditions becomes apparent (Fig. 5). A predictive model of alternating high/low climate variability, based on the modulation of precession by eccentricity, captures many of these complexities for East Africa. The model highlights the numerous instances when arid-moist oscillations became magnified following eras of relative stability. Heightened variability in precipitation intensified the unpredictability of resources and other ecological parameters confronted by the biota, including hominins. Eastern Mediterranean sapropels appear to provide the best existing record that mirrors the high/low variability model, although lakes (Trauth et al., 2007, 2010), dust records (deMenocal, 2004, 2011), and plant biomarker $\delta^{13}C$ (Feakins et al., 2005) also provide evidence of alternating intervals of East African climate variability over time. The observation forwarded here, consistent with the mechanisms delineated by Trauth et al. (2010), is that both aridity and high moisture were expressed in the high variability intervals, and this observation has now been documented at Olduvai (Ashley, 2007), East Turkana (Lepre et al., 2007), Hadar (Campisano and Feibel, 2007), Olorgesailie (Behrensmeyer et al., 2002; see Fig. 4), and in other Pliocene and Pleistocene sedimentary basins of East Africa.

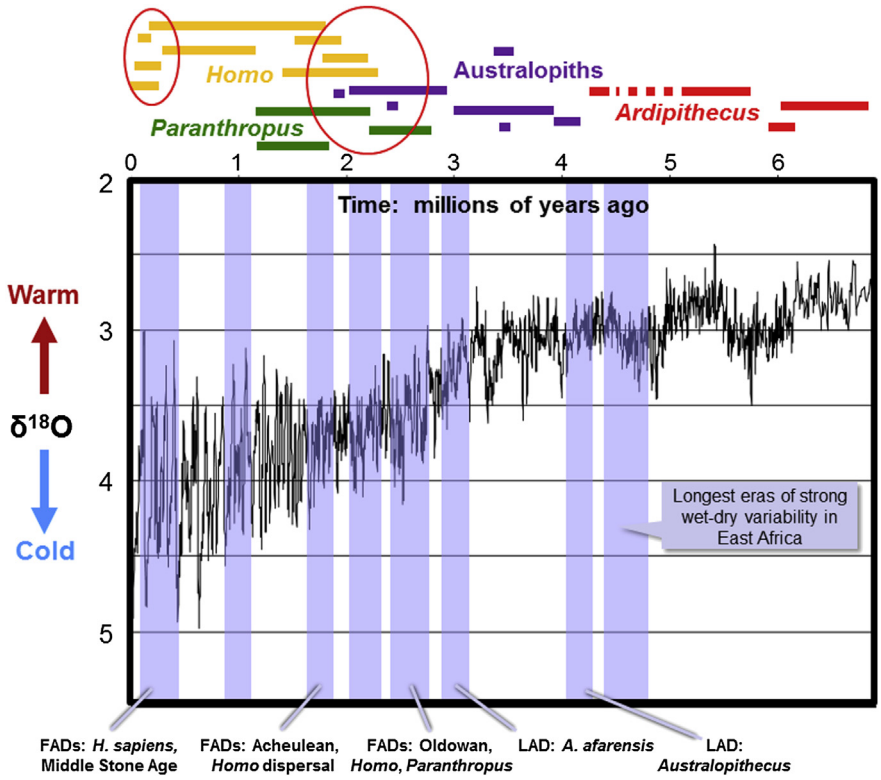


Fig. 5. A synthesis of paleoclimate and human evolution over the past 7 million years. Oxygen isotope data (Zachos et al., 2001) depict the overall cooling trend leading to the intensification of Northern Hemisphere glacial cycles during the Pliocene, along with the increasing range of variability over the period of hominin evolution. The genus *Homo* evolved as global climate variability increased, and *H. sapiens* emerged during the highest amplitude variations. For East Africa, eight intervals of prolonged (duration >190 kyr) strong wet–dry climate variability (blue vertical bars) are also shown, along with associated key events in African hominin evolution (bottom). Approximate species ranges for hominin taxa (top) are shown as colored horizontal bars (red: *Ardipithecus* and other Mio-Pliocene lineages; purple: lineages assigned to *Australopithecus*; green: *Paranthropus*; yellow: *Homo*). Peaks in hominin species diversity and turnover (red ovals) correspond to the closely-spaced intervals of predicted high climate variability between ~2.8 and 1.7 Ma, and the prolonged one between ~360 and 50 ka.

An advantage afforded by this high/low variability framework is that it defines a series of recurrent shifts in climate oscillation throughout the past several million years, enabling an opportunity for repeated observations and analysis of the associations between environmental and evolutionary change. The synthesis of environmental and paleoanthropological data presented here suggests a consistent association of critical evolutionary events with prolonged periods of strong environmental instability over the course of hominin evolution. Evolutionary explanations that emphasize specific habitat reconstructions (e.g., C_4 grassland) or stability (e.g., persistent warmth) lack support, while those that focus on adaptive versatility in response to environmental variability are gaining support.

This finding favors the variability selection explanation of many events in hominin evolution. This explanation implies that the adaptations that unpin much of human environmental interactions reflect past evolutionary responses to ecological instability and environmental novelty. This conclusion is tentative, however, in that much remains to be done to develop a coherent study of environmental-evolutionary linkages. In particular, different animals have responded to climate and landscape change in different ways, a point Vrba (1980) noted in how impala and aardvarks, on the one hand, and alcelaphine bovids, on the other, have greatly differed in their degree of morphological change and speciation in reaction to the environmental history of East Africa. Much remains to be learned about how early African hominins fit into the wide range of possible responses. Links in the causal chain from climate dynamics to local and regional resource landscapes and to the different ways in which organisms responded still need to be developed. Only then can we expect to improve the answer to the question of how climate has influenced human evolution.

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References

Ashley, G.M., 2007. Orbital rhythms, monsoons, and playa lake response, Olduvai Basin, equatorial East Africa (ca. 1.85–1.74 Ma). Geology 35, 1091–1094.
Bailey, G.N., Reynolds, S.C., King, G.C.P., 2011. Landscapes of human evolution: models and methods of tectonic geomorphology and the reconstruction of hominin landscapes. Journal of Human Evolution 60, 257–280.
Bartholomew, G.A., Birdsell, J.B., 1953. Ecology and the protohominids. American Anthropologist 55, 481–498.
Behrensmeyer, A.K., Hook, R.W., 1992. Paleoenvironmental contexts and taphonomic modes. In: Behrensmeyer, A.K., Damuth, J.D., DiMichele, W.A., Potts, R.,

Sues, H.-D., Wing, S.L. (Eds.), Terrestrial Ecosystems Through Time. University of Chicago, Chicago, pp. 15–136.
Behrensmeyer, A.K., Todd, N.E., Potts, R., McBrinn, G.E., 1997. Late Pliocene faunal turnover in the Turkana Basin, Kenya and Ethiopia. Science 278, 1589–1594.
Behrensmeyer, A.K., Potts, R., Deino, A., Ditchfield, P., 2002. Olorgesailie, Kenya: a million years in the life of a rift basin. In: Renaut, R.W., Ashley, G.M. (Eds.), Sedimentation in Continental Rifts. SEPM Special Publication 73, pp. 97–106.
Bobe, R., Alemseged, Z., Behrensmeyer, A.K. (Eds.), 2007. Hominin Environments in the East African Pliocene: an Assessment of the Faunal Evidence. Springer, New York.
Bobe, R., Behrensmeyer, A.K., 2004. The expansion of grassland ecosystems in Africa in relation to mammalian evolution and the origin of the genus *Homo*. Palaeogeography, Palaeoclimatology, Palaeoecology 207, 399–420.
Bobe, R., Eck, G.G., 2001. Responses of African Bovids to Pliocene climatic change. Paleobiology 27 (Suppl. 2), 1–47.
Bobe, R., Leakey, M.G., 2009. Ecology of Plio-Pleistocene mammals in the Omo-Turkana Basin and the emergence of *Homo*. In: Grine, F.E., Fleagle, J.G., Leakey, R.E. (Eds.), The First Humans: Origins of the Genus *Homo*. Springer, New York, pp. 173–184.
Bonnefille, R., 1995. A reassessment of the Plio-Pleistocene pollen record of East Africa. In: Vrba, E.S., Denton, G.H., Partridge, T.C., Burkle, L.H. (Eds.), Paleoclimate and Evolution, with Emphasis on Human Origins. Yale University Press, New Haven, Connecticut, pp. 299–310.
Bonnefille, R., 2010. Cenozoic vegetation, climate changes and hominid evolution in tropical Africa. Global and Planetary Change 72, 390–411.
Bonnefille, R., Potts, R., Chalié, F., Jolly, D., Peyron, O., 2004. High-resolution vegetation and climate change associated with Pliocene *Australopithecus afarensis*. Proceedings of the National Academy of Sciences United States of America 101, 12125–12129.
Brooks, A., Behrensmeyer, K., Yellen, J., Deino, A., Sharp, W., Potts, R., 2007. After the Acheulean: stratigraphy, dating, and archeology of two new formations in the Olorgesailie basin, southern Kenya Rift. Paleoanthropology 2007, A5.
Campisano, C.J., Feibel, C.S., 2007. Connecting local environmental sequences to global climate patterns: evidence from the hominin-bearing Hadar Formation, Ethiopia. Journal of Human Evolution 53, 515–527.
Cerling, T.E., 1992. Development of grasslands and savannas in East Africa during the Neogene. Palaeogeography, Palaeoclimatology, Palaeoecology 97, 241–247.
Cerling, T.E., Hay, R.L., 1986. An isotopic study of paleosol carbonates from Olduvai Gorge. Quaternary Research 25, 63–78.
Cerling, T.E., Wynn, J.G., Andanje, S.A., Bird, M.I., Korir, D.K., Levin, N.E., Mace, W., Macharia, A.N., Quade, J., Remien, C.H., 2011. Woody cover and hominin environments in the past 6 million years. Nature 476, 51–56.
Cohen, A.S., Stone, J.R., Beuning, K.R.M., Park, L.E., Reinthal, P.N., Dettman, D., Scholz, C.A., Johnson, T.C., King, J.W., Talbot, M.R., Brown, E.T., Ivory, S.J., 2007. Ecological consequences of early Late Pleistocene megadroughts in tropical Africa. Proceedings of the National Academy of Sciences United States of America 104, 16422–16427.
Darwin, C., 1871. The Descent of Man and Selection in Relation to Sex. Murray, London.
Deino, A., Potts, R., 1990. Single crystal $^{40}Ar/^{39}Ar$ dating of the Olorgesailie Formation, southern Kenya rift. Journal of Geophysical Research 95 (B6), 8453–8470.
Deino, A.L., Kingston, J.D., Glen, J.M., Edgar, R.K., Hill, A., 2006. Precessional forcing of lacustrine sedimentation in the late Cenozoic Chemeron Basin, Central Kenya Rift. Earth and Planetary Science Letters 247, 41–60.
deMenocal, P.B., 1995. Plio-Pleistocene African climate. Science 270, 53–59.
deMenocal, P.B., 2004. African climate change and faunal evolution during the Pliocene–Pleistocene. Earth and Planetary Science Letters 220, 3–24.
deMenocal, P.B., 2011. Climate and human evolution. Science 331, 540–541.
deMenocal, P.B., Bloemendal, J., 1995. Plio-Pleistocene climatic variability in subtropical Africa and the paleoenvironment of hominid evolution. In: Vrba, E.S., Denton, G.H., Partridge, T.C., Burkle, L.H. (Eds.), Paleoclimate and Evolution, with Emphasis on Human Origins. Yale University Press, New Haven, Connecticut, pp. 262–288.
Domínguez-Rodrigo, M., Pickering, T.R., Semaw, S., Rogers, M.J., 2005. Cutmarked bones from Pliocene archaeological sites at Gona, Ethiopia: implications for the function of the world's oldest stone tools. Journal of Human Evolution 48, 109–121.
Donges, J.F., Donner, R.V., Trauth, M.H., Marwan, N., Schellnhuber, H.-J., Kurths, J., 2011. Nonlinear detection of paleoclimate-variability transitions possibly related to human evolution. Proceedings of the National Academy of Sciences United States of America 108, 20423–20428.
Emeis, K.-C., Sakamoto, T., Wehausen, R., Brumsack, H.-J., 2000. The sapropel record of the eastern Mediterranean Sea – results of Ocean Drilling Program Leg 160. Palaeogeography, Palaeoclimatology, Palaeoecology 158, 371–395.
Faith, J.T., Potts, R., Plummer, T.W., Bishop, L.C., Marean, C.W., Tryon, C.A., 2012. New perspectives on middle Pleistocene change in the large mammal faunas of East Africa: *Damaliscus hypsodon* sp. nov. (Mammalia, Artiodactyla) from Lainyamok, Kenya. Palaeogeography, Palaeoclimatology, Palaeoecology 361–362, 84–93.
Feakins, S.J., deMenocal, P.B., Eglinton, T.I., 2005. Biomarker records of late Neogene changes in northeast African vegetation. Geology 33, 977–980.
Feakins, S.J., Levin, N.E., Liddy, H.M., Sieracki, A., Eglinton, T.I., Bonnefille, R., 2013. Northeast African vegetation change over 12 million years. Geology 41, 295–298.
Finlayson, C., Pacheco, F.G., Rodríguez-Vidal, J., Fa, D.A., Gutierrez López, J.M., et al., 2006. Late survival of Neanderthals at the southernmost extreme of Europe. Nature 443, 850–853.
Guiot, J.A., Pons, J.L., de Beaulieu, J.L., Reille, M., 1989. A 140,000-year continental climate reconstruction from two European pollen records. Nature 338, 309–313.
Green, R.E., Krause, J., Briggs, A.W., Maricic, T., Stenzel, U., Kircher, M., Patterson, N., et al., 2010. A draft sequence of the Neanderthal genome. Science 328, 710–722.
Grove, M., 2011a. Speciation, diversity, and Mode 1 technologies: the impact of variability selection. Journal of Human Evolution 61, 306–319.
Grove, M., 2011b. Change and variability in Plio-Pleistocene climates: modelling the hominin response. Journal of Archaeological Science 38, 3038–3047.
Hilgen, F.J., Abdul Aziz, H., Krijgsman, W., Langereis, C.G., Lourens, L.J., Meulenkamp, J.E., Raffi, I., Steenbrink, J., Turco, E., van Vugt, N., Wijbrans, J.R., Zachariasse, W.J., 1999. Present status of the astronomical (polarity) time-scale for the Mediterranean Late Neogene. Philosophical Transactions: Mathematical, Physical and Engineering Sciences 357, 1931–1947.
Hopley, P., Maslin, M.A., 2010. Climate-averaging of terrestrial faunas – an example from the Plio-Pleistocene of South Africa. Paleobiology 36, 32–50.
Isaac, G. Ll, 1977. Olorgesailie: Archaeological Studies of a Middle Pleistocene Lake Basin in Kenya. Chicago University Press, Chicago.
Joordens, J.C.A., Vonhof, H.B., Feibel, C.S., Lourens, L.J., Dupont-Nivet, G., van der Lubbe, J.H.J.L., Sier, M.J., Davies, G.R., Kroon, D., 2011. An astronomically-tuned climate framework for hominins in the Turkana Basin. Earth and Planetary Science Letters 307, 1–8.
Kingston, J.D., 2007. Shifting adaptive landscapes: progress and challenges in reconstructing early hominid environments. Yearbook of Physical Anthropology 50, 20–58.
Kingston, J.D., Deino, A.L., Edgar, R.K., Hill, A., 2007. Astronomically forced climate change in the Kenyan Rift Valley 2.7–2.55 Ma: implications for the evolution of early hominin ecosystems. Journal of Human Evolution 53, 487–503.
Klein, R.G., 1989. The Human Career: Human Biological and Cultural Origins. University of Chicago, Chicago.
Laland, K., Odling-Smee, J., Feldman, L., 2001. Cultural niche construction and human evolution. Journal of Evolutionary Biology 14, 22–33.
Leakey, M.G., Feibel, C.S., McDougall, I., Ward, C., Walker, A., 1998. New specimens and confirmation of an early age for *Australopithecus anamensis*. Nature 393, 62–66.
Laporte, L.F., Zihlman, A.L., 1983. Plates, climate and hominoid evolution. South African Journal Science 79, 96–110.
Lepre, C.J., Quinn, R.L., Joordens, J.C.A., Swisher III, C.C., Feibel, C.S., 2007. Plio-Pleistocene facies environments from the KBS Member, Koobi Fora Formation: implications for climate controls on the development of lake-margin hominin habitats in the northeast Turkana Basin (northwest Kenya). Journal of Human Evolution 53, 504–514.
Levin, N.E., Quade, J., Simpson, S.W., Semaw, S., Rogers, M., 2004. Isotopic evidence for Plio-Pleistocene environmental change at Gona, Ethiopia. Earth and Planetary Science Letters 219, 93–110.
Levin, N.E., Brown, F.H., Behrensmeyer, A.K., Bobe, R., Cerling, T.E., 2011. Paleosol carbonates from the Omo Group: isotopic records of local and regional environmental change in East Africa. Palaeogeography, Palaeoclimatology, Palaeoecology 307, 75–89.
Lisiecki, L.E., Raymo, M.E., 2005. A Pliocene–Pleistocene stack of 57 globally distributed benthic delta O-18 records. Paleoceanography 20 (PA1003), 1–17.
Lisiecki, L.E., Raymo, M.E., 2007. Plio-Pleistocene climate evolution: trends and transitions in glacial cycle dynamics. Quaternary Science Reviews 26, 56–69.
Lourens, L.J., Antonarakou, A., Hilgen, F., Van Hoof, A.A.M., Vergnaud-Grazzini, C., Zachariasse, W.J., 1996. Evaluation of the Plio-Pleistocene astronomical timescale. Paleoceanography 11, 391–413.
Maslin, M.A., Christensen, B., 2007. Tectonics, orbital forcing, global climate change, and human evolution in Africa. Journal of Human Evolution 53, 443–464.
Maslin, M.A., Trauth, M.H., 2009. Plio-Pleistocene East African pulsed climate variability and its influence on early human evolution. In: Grine, F.E., Leakey, R.E., Fleagle, J.G. (Eds.), The First Humans: Origins of the Genus *Homo*. Springer, New York, pp. 151–158.
Mellars, P., 2004. Neanderthals and the modern human colonization of Europe. Nature 432, 461–465.
Meyers, S.R., Hinnov, L.A., 2010. Northern Hemisphere glaciation and the evolution of Plio-Pleistocene climate noise. Paleoceanography 25 (PA3207), 1–11.
National Research Council, 2010. Understanding Climate's Influence on Human Evolution. National Academies Press, Washington, DC.
Navarrete, A., van Schaik, C.P., Isler, K., 2011. Energetics and the evolution of human brain size: no gut-brain trade-off in mammals. Nature 480, 91–93.
Odling-Smee, J., Laland, K., Feldman, M., 2003. Niche Construction. In: Monographs in Population Biology, No. 37. Princeton University Press, Princeton, New Jersey.
Owen, R.B., Potts, R., Behrensmeyer, A.K., Ditchfield, P., 2008. Diatomaceous sediments and environmental change in the Pleistocene Olorgesailie Formation, southern Kenya Rift Valley. Palaeogeography, Palaeoclimatology, Palaeoecology 269, 17–37.
Owen, R.B., Renaut, R.W., Potts, R., Behrensmeyer, A.K., 2011. Geochemical trends through time and lateral variability of diatom floras in the Pleistocene Olorgesailie Formation, southern Kenya Rift Valley. Quaternary Research 76, 167–179.
Passey, B.H., Levin, N.E., Cerling, T.E., Brown, F.H., Eiler, J.M., 2010. High-temperature environments of human evolution in East Africa based on bond ordering in

paleosol carbonates. Proceedings of the National Academy of Sciences United States of America 107, 11,245–11,249.
Potts, R., 1989. Olorgesailie: new excavations and findings in Early and Middle Pleistocene contexts, southern Kenya rift valley. Journal of Human Evolution 18, 477–484.
Potts, R., 1994. Variables vs. models of early Pleistocene hominid land use. Journal of Human Evolution 27, 7–24.
Potts, R., 1996a. Humanity's Descent: the Consequences of Ecological Instability. William Morrow, New York.
Potts, R., 1996b. Evolution and climate variability. Science 273, 922–923.
Potts, R., 1998a. Variability selection in hominid evolution. Evolutionary Anthropology 7, 81–96.
Potts, R., 1998b. Environmental hypotheses of hominin evolution. Yearbook of Physical Anthropology 41, 93–136.
Potts, R., 2001. Mid-Pleistocene environmental change and human evolution. In: Barham, L., Robson-Brown, K. (Eds.), Human Roots: Africa and Asia in the Middle Pleistocene. Western Academic Press, Bristol, pp. 5–21.
Potts, R., 2002. Complexity and adaptability in human evolution. In: Goodman, M., Moffat, A. (Eds.), Probing Human Origins. American Academy of Arts and Sciences, Cambridge, Massachusetts, pp. 33–57.
Potts, R., 2007. Environmental hypotheses of Pliocene human evolution. In: Bobe, R., Alemseged, Z., Behrensmeyer, A.K. (Eds.), Hominin Environments in the East African Pliocene: an Assessment of the Faunal Evidence. Springer, New York, pp. 25–50.
Potts, R., 2012a. Evolution and environmental change in early human prehistory. Annual Review of Anthropology 41, 151–167.
Potts, R., 2012b. Environmental and behavioral evidence pertaining to the evolution of early Homo. Current Anthropology 53 (Suppl. 6), S299–S317.
Potts, R., Deino, A., 1995. Mid-Pleistocene change in large mammal faunas of the southern Kenya rift. Quaternary Research 43, 106–113.
Potts, R., Behrensmeyer, A.K., Ditchfield, P., 1999. Paleolandscape variation in early Pleistocene hominid activities: members 1 and 7, Olorgesailie Formation, Kenya. Journal of Human Evolution 37, 747–788.
Potts, R., Behrensmeyer, A.K., Deino, A., Ditchfield, D., Clark, J., 2004. Small mid-Pleistocene hominin associated with East African Acheulean technology. Science 305, 75–78.
Prentice, M.L., Denton, G.H., 1988. The deep-sea oxygen isotope record, the global ice sheet system and hominid evolution. In: Grine, F.E. (Ed.), Evolutionary History of the 'Robust' Australopithecines. Aldine de Gruyter, New York, pp. 383–403.
Quinn, R.L., Lepre, C.J., Wright, J.D., Feibel, C.S., 2007. Paleogeographic variations of pedogenic carbonate $\delta^{13}C$ values from Koobi Fora, Kenya: implications for floral compositions of Plio-Pleistocene hominin environments. Journal of Human Evolution 53, 560–573.
Reed, K.E., 1997. Early hominid evolution and ecological change through the African Plio-Pleistocene. Journal of Human Evolution 32, 289–322.
Rogers, A.R., Iltis, D., Wooding, S., 2004. Genetic variation at the MC1R locus and the time since loss of human body hair. Current Anthropology 45, 105–108.
Rossignol-Strick, M., 1983. African monsoons, an immediate climate response to orbital insolation. Nature 304, 46–49.
Scholz, C.A., Johnson, T.C., Cohen, A.S., King, J.W., Peck, J.A., Overpeck, J.T., Talbot, M.R., Brown, E.T., Kalindekafe, L., Amoako, P.Y.O., Lyons, R.P., Shanahan, T.M., Castañeda, I.S., Heil, C.W., Forman, S.L., McHargue, L.R., Beuning, K.R., Gomez, J., Pierson, J., 2007. East African megadroughts between 135 and 75 thousand years ago and bearing on early-modern human origins. Proceedings of the National Academy of Sciences United States of America 104, 16416–16421.
Semaw, S., Rogers, M.J., Quade, J., Renne, P.R., Butler, R.F., Dominguez-Rodrigo, M., Stout, D., Hart, W.S., Pickering, T., Simpson, S.W., 2003. 2.6-Million-year-old stone tools and associated bones from OGS-6 and OGS-7, Gona, Afar, Ethiopia. Journal of Human Evolution 45, 169–177.
Shackleton, N.J., 1995. New data on the evolution of Pliocene climatic variability. In: Vrba, E.S., Denton, G.H., Partridge, T.C., Burckle, L.H. (Eds.), Paleoclimate and Evolution with Emphasis on Human Origins. Yale University Press, New Haven, Connecticut, pp. 242–248.
Tauxe, L., Deino, A.L., Behrensmeyer, A.K., Potts, R., 1992. Pinning down the Brunhes/Matuyama and upper Jaramillo boundaries: a reconciliation of orbital and isotopic time scales. Earth and Planetary Science Letters 109, 561–572.
Trauth, M.H., Maslin, M.A., Deino, A.L., Strecker, M.R., 2005. Late Cenozoic moisture history of East Africa. Science 309, 2051–2053.
Trauth, M.H., Maslin, M.A., Deino, A.L., Strecker, M.R., Bergner, A.G.N., Dühnforth, M., 2007. High- and low-latitude forcing of Plio-Pleistocene East African climate and human evolution. Journal of Human Evolution 53, 475–486.
Trauth, M.H., Larrasoaña, J.C., Mudelsee, M., 2009. Trends, rhythms and events in Plio-Pleistocene African climate. Quaternary Science Reviews 28, 399–411.
Trauth, M.H., Maslin, M.A., Deino, A.L., Junginger, A., Lesoloyiae, M., Odada, E.O., Olago, D.O., Olaka, L.A., Strecker, M.R., Tiedemann, R., 2010. Human evolution in a variable environment: the amplifier lakes of Eastern Africa. Quaternary Science Reviews 29, 2981–2988.
van Andel, T.H., 2003. Glacial environments I: the Weichselian climate in Europe between the end of the OIS-5 interglacial and the Last Glacial Maximum. In: van Andel, T.H., Davies, W.D. (Eds.), Neanderthals and Modern Humans in the European Landscape of the Last Glaciation – Archaeological Results of the Stage 3 Project. ed. McDonald Institute for Archaeological Research, Cambridge, U.K, pp. 9–19.
Vrba, E.S., 1980. Evolution, species, and fossils: how does life evolve? South African Journal of Science 76, 61–84.
Vrba, E.S., 1985a. Environment and evolution: alternative causes of the temporal distribution of evolutionary events. South African Journal of Science 81, 229–236.
Vrba, E.S., 1985b. Ecological and adaptive changes associated with early hominid evolution. In: Delson, E. (Ed.), Ancestors: the Hard Evidence. Liss, NY, pp. 63–71.
Vrba, E.S., 1988. Late Pliocene climatic events and hominid evolution. In: Grine, F.E. (Ed.), Evolutionary History of the 'Robust' Australopithecines. Aldine de Gruyter, New York, pp. 405–426.
Vrba, E.S., 1992. Mammals as a key to evolutionary theory. Journal of Mammalogy 73, 1–28.
Vrba, E.S., 1994. An hypothesis of heterochrony in response to climatic cooling and its relevance to early hominid evolution. In: Corruccini, R., Ciochon, R.L. (Eds.), Integrative Paths to the Past. Prentice Hall, Englewood Cliffs, New Jersey, pp. 345–376.
Vrba, E.S., 1995a. On the connections between paleoclimate and evolution. In: Vrba, E.S., Denton, G.H., Partridge, T.C., Burckle, L.H. (Eds.), Paleoclimate and Evolution with Emphasis on Human Origins. Yale University Press, New Haven, Connecticut, pp. 24–45.
Vrba, E.S., 1995b. The fossil record of African antelopes (Mammalia, Bovidae) in relation to human evolution and paleoclimate. In: Vrba, E.S., Denton, G.H., Partridge, T.C., Burckle, L.H. (Eds.), Paleoclimate and Evolution with Emphasis on Human Origins. Yale University Press, New Haven, Connecticut, pp. 385–424.
Vrba, E.S., Denton, G.H., Prentice, M.L., 1989. Climatic influences on early hominid behavior. Ossa 14, 127–156.
Vrba, E.S., Denton, G.H., Partridge, T.C., Burckle, L.H. (Eds.), 1995. Paleoclimate and Evolution with Emphasis on Human Origins. Yale University Press, New Haven, Connecticut.
Washburn, S.L., 1960. Tools and human evolution. Scientific American 203 (3), 63–75.
Werdelin, L., Sanders, W.J. (Eds.), 2010. Cenozoic Mammals of Africa. University of California Press, Berkeley, California.
White, T.D., WoldeGabriel, G., Asfaw, B., Ambrose, S., Beyene, Y., Bernor, R.L., Boisserie, J.-R., Currie, B., Gilbert, H., Haile-Selassie, Y., Hart, W.K., Hlusko, L.J., Howell, F.C., Kono, R.T., Lehmann, T., Louchart, A., Lovejoy, C.O., Renne, P.R., Saegusa, H., Vrba, E.S., Wesselman, H., Suwa, G., 2006. Asa Issie, Aramis and the origin of *Australopithecus*. Nature 440, 883–889.
Wilson, E.O., 1978. On Human Nature. Harvard University Press, Cambridge, Massachusetts.
Wolpoff, M.H., 1980. Paleoanthropology. Knopf, New York.
Wynn, J.G., 2004. Influence of Plio-Pleistocene aridification on human evolution: evidence from paleosols of the Turkana Basin, Kenya. American Journal of Physical Anthropology 123, 106–118.
Zachos, J., Pagani, M., Sloan, L., Thomas, E., Billups, K., 2001. Trends, rhythms, and abberations in global climate 65 Ma to present. Science 292, 686–693.